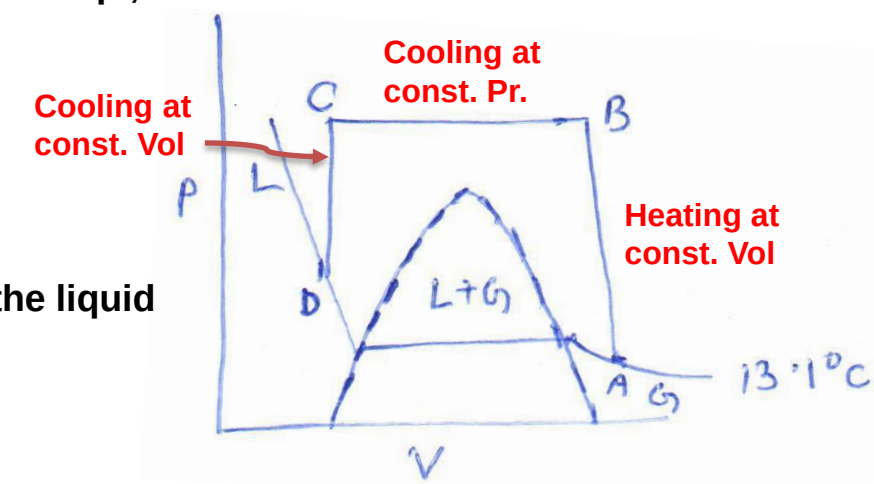
Continuity of States

In the figure, A and D lie on the same isotherm at 13.1 ^{o}C temp., below the T_c of CO_2.

Point A → gaseous state
Point D → liquid state
Dashed area → contains liquid-gas in equilibrium

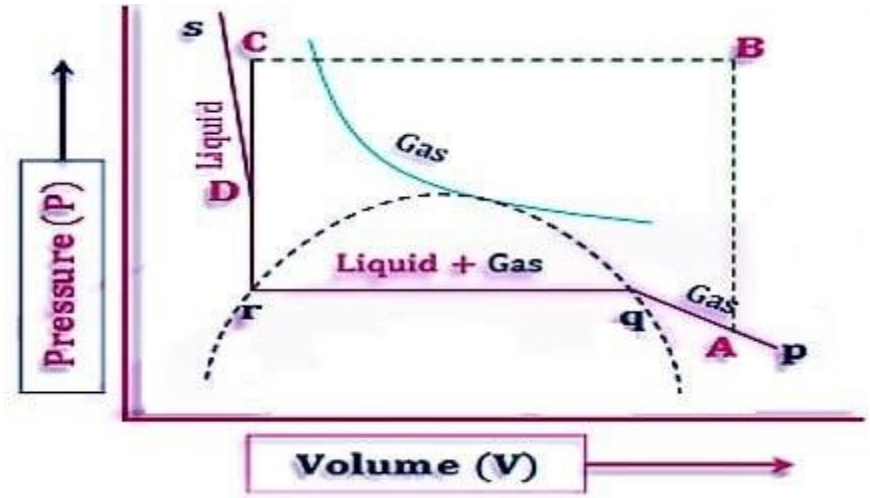
It is possible to shift from the gaseous state point A to the liquid state point D continuously without passing through the discontinuous dashed area.

Two-phase region and continuity of states



- ✓ Let the gas at the state point A is heated to B at constant vol. along AB.
- ✓ Then, the gas is gradually cooled at constant pressure along BC, the vol. is reduced considerably.
- ✓ The gas is again cooled at constant vol. until the state point D is reached.
- ✓ No where in the process liquid would appear. At D, the system is highly compressed gas. But the curve shows that this state point is for the liquid state.
- ✓ Thus, there is hardly any difference between the liquid state and the gaseous state and there is no line of demarcation between the two phases. This is the continuity of states.
- ✓ The point D we may refer as liquid state or highly compressed gaseous state. In the absence of discontinuity there is no fundamental way to distinguish liquid or gas.
- ✓ The gas is continuously transferred to the liquid without passing the usual process of condensation.

Continuity of States



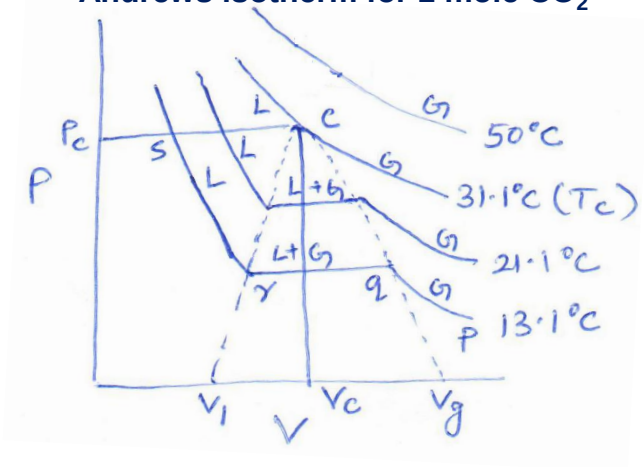
- ✓ At the critical temperature, the boundary between the liquid phase and gas phase disappears.
- ✓ The densities of the substance in the two phases are the same.
- ✓ Continuity of state is the property of a transition between two states of matter, as between gas and liquid, during which there are no abrupt changes in physical properties.
- ✓ Thus, the transformation from the gaseous state to the liquid state or vice versa may be made gradually, without passing through the phenomenon of condensation.
- ✓ Nowhere during the transformation two phases will co-exist.

Continuity of States

What is the physical significance of the volume range being widened with decreasing pressure??

Answer: You are changing temp.
Physical significance: You need higher saturation Pr., so the range of vol. over which condensation occurs is smaller.

Andrews isotherm for 1 mole CO_2

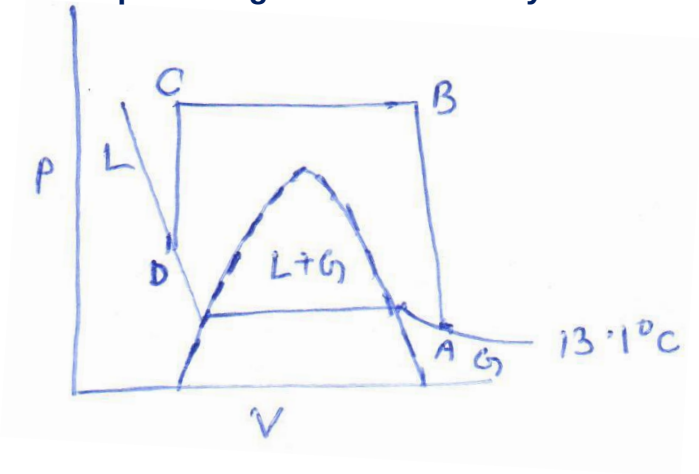


How can liquid and gaseous states be the same?

Answer:
They are not same. There is no line of demarcation between the two phases. **This is the continuity of states.**

In the absence of discontinuity there is no fundamental way to distinguish liquid or gas. The gas is continuously transferred to the liquid without passing the usual process of condensation.

Two-phase region and continuity of states



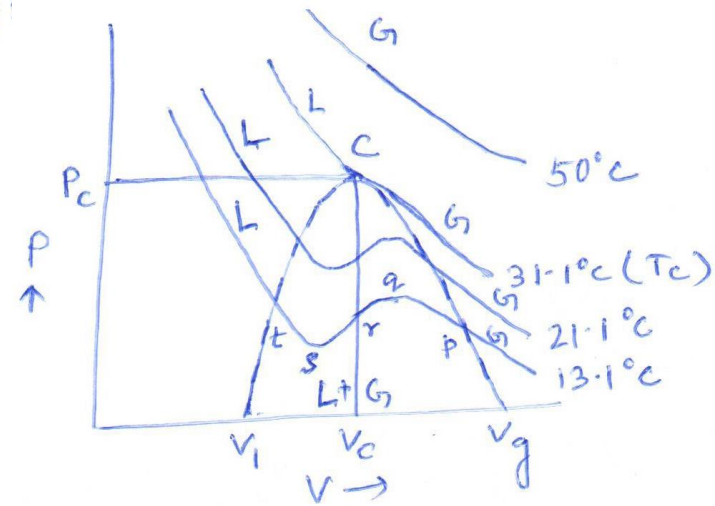
Question

Inside Jupiter, liquid hydrogen is found. But the temperature of Jupiter is extremely high and definitely above the critical temperature of hydrogen. So how is liquid hydrogen found in Jupiter?

Critical Phenomena and Vander Waals Equation

The VDW Equn: for 1 mole gas : $P = \frac{RT}{V-b} - \frac{a}{V}$

VDW Isotherm for 1 mole CO_2



T_c = 31.1 °C; P_c = 72.9 atm and V_c = 94.2 cc/mol

Taking the values of *a* and *b* for CO_2, it is possible to collect data *P* and *V* at 13.1, 21.1, 31.1 and 50 °C, and draw the similar isotherms like that of Andrews.

→ Vander Waals isotherms

The experimental Andrews isotherms coincide with the Vander Waals isotherms in all the points except the region where the gas and liquid are co-existing.

The horizontal lines are replaced by wavy lines. These wavy portion of Vander Waals curves poses two limitations of the VDWE which are not realized in practice.

The isotherm *pqrst* shows that there are three volumes of the system at a give *T* and *P*.

Again at the portion *srq*, it shows that with increase of Pr., vol. is also increased.

These two points show the limitations of VDWE.

with increase of temperature, the minimum and maximum points come close to each other and at the critical point, both coalesce. The slope and curvature both are zero at the point. That is

$$\left(\frac{\partial P}{\partial V}\right)_T = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$$

Relation between critical constants (P_c, V_c and T_c) and Vander Walls constants (a and b)

Differentiating VDWE, $P = \frac{RT}{V-b} - \frac{a}{V^2}$ with respect to V at constant T, we get the slope, $\left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3}$

and the curvature, $\left(\frac{\partial^2 P}{\partial V^2}\right)_T = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4}$

But, at the critical state, $\left(\frac{\partial P}{\partial V}\right)_T = 0$ and $\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$,

and $T = T_c$, $V = V_c$

So, $\frac{RT_c}{(V_c-b)^2} = \frac{2a}{V_c^3}$ and $\frac{2RT_c}{(V_c-b)^3} = \frac{6a}{V_c^4}$

Solving the two equations, we get,

$V_c = 3b$ and $T_c = \frac{8a}{27Rb}$

Again, VDWE at the critical point, $P_c = \frac{RT_c}{V_c-b} - \frac{a}{V_c^2}$

Inserting the value of T_c and V_c, we have, $P_c = \frac{a}{27b^2}$

Thus, the relation between critical constants and VDW const.s are

$V_c = 3b$, $T_c = \frac{8a}{27Rb}$, and $P_c = \frac{a}{27b^2}$

Relation between critical constants (P_c, V_c and T_c) and Vander Walls constants (a and b)

One property of the gas is critical coefficient, $\frac{RT_c}{P_cV_c}$

Putting V_c, P_c and T_c, $\frac{RT_c}{P_cV_c} = \frac{8}{3} = 2.66$.

Thus, critical coefficient is predicted constant for all VDW gases.

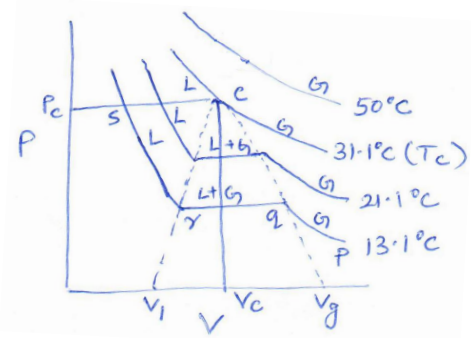
The value of compressibility factor at the critical state,

$$Z_c = \frac{P_cV_c}{RT_c} = \frac{3}{8}$$ and it is less than 1.

Thus, at the critical state, the gas is more compressible.

These values are tested experimentally whether the critical coefficient is constant and = 2.66 for all gases but it is found that it varies from gas to gas and the average value is about 3.66.

Applications: Calculations of Vander Waals constant *a* and *b*.



$b = \frac{V_c}{3}$; V_c is avoided in the relation as it is not easily determined experimentally.

V_c is replaced by using the value of critical coefficient,

$$V_c = \frac{3}{8}\frac{RT_c}{P_c}$$

We know,

$V_c = 3b$, $T_c = \frac{8a}{27Rb}$, and $P_c = \frac{a}{27b^2}$

Therefore, VDW constants are, $b = \frac{1}{8}\frac{RT_c}{P_c}$

and $a = \frac{27}{64}\frac{R^2T_c^2}{P_c}$

The critical constants of a gas can be determined experimentally and so the values of the VDW constants *a* and *b* can be calculated.

Reduced form of Vander Waals Equation of State

VDWE can be expressed in terms of their reduced variables in stead of *T*, *P* and *V*. The reduced variables are defined as the actual variables divided by the corresponding critical constants. Thus,

reduced Pr., $\pi = \frac{P}{P_c}$

reduced Temp., $\theta = \frac{T}{T_c}$

reduced vol. $\phi = \frac{V}{V_c}$

Replacing *P*, *V* and *T* in the VDWE by corresponding reduced variables, we have

$$\left[\pi P_c + \frac{a}{(\phi V_c)^2}\right](\phi V_c - b) = R\theta T_c$$

Inserting the value of critical const.s, $V_c = 3b$, $T_c = \frac{8a}{27Rb}$ and $P_c = \frac{a}{27b^2}$ in the above eqn., we get the reduced form of VDWE : $\left(\pi + \frac{3}{\phi^2}\right)(3\phi - 1) = 8\theta$

- ✓ **The important feature of the equation is that it is independent of VDW constants, *a* and *b*.**
- ✓ **Thus, it appears that the equation is independent of the nature of the gas and general as it is in ideal gas equation.**
- ✓ **But it is not true. The characteristic gas constants now remain disguise within the reduced variables (π, ϕ and θ) as these variables contain critical constants (P_c, V_c and T_c) and these are characteristic constants of the gas.**
- ✓ **Thus, it is not that generality lost in VDWE is regained in this form.**
- ✓ **The reduced equation of state gives birth one important generalization, called law of corresponding states.**

Limitations of Vander Waals Equation

① The equn. predicts $V_c = 3b$ but the mean value is $V_c = 2b$

② VDWE suggests, $\frac{RT_c}{P_c V_c} = \frac{8}{3} = 2.66$, but the average value is 3.66.

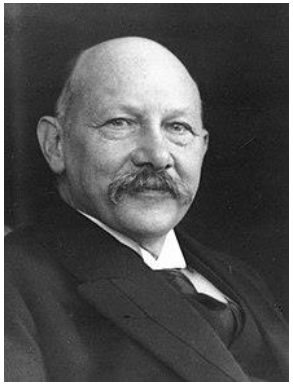
③ $\frac{T_B}{T_c} = \frac{27}{8} = 3.375$, but the value is 2.98.

So, we can conclude that VDWE explains the behaviour of the real gas qualitatively but it fails to explain quantitatively.

Kammerlingh - Onnes Virial Equation:

The equation of state of a gas can be satisfactorily expressed as a power series of Vol. or Pr., suggested by H. Kammerlingh-Onnes in 1901 as;

$$PV = RT\left[1 + \frac{B}{V} + \frac{C}{V^2} + \frac{D}{V^3} + \cdots\right]$$



Nobel Prize in Physics (1913)

B = 2nd virial coefficient and is most important in the expression.

B, C, D is the virial coefficients and dependent on T and on the nature of the gas, their values are small in magnitude.

Virial equn. as power series of P is given by

$$PV = RT\left[1 + B_1 P + C_1 P^2 + D_1 P^3 + \cdots\right]$$

Kammerlingh-Onnes Virial Equation

VDWE of real gas can be recast in virial form as follows;

$$P = \frac{RT}{V-b} - \frac{a}{V^2} \Rightarrow PV = \frac{RTV}{V-b} - \frac{a}{V}$$

$$\Rightarrow PV = \frac{RT}{\left(1-\frac{b}{V}\right)} - \frac{a}{V}$$

$$= RT\left(1-\frac{b}{V}\right)^{-1} - \frac{a}{V}$$

$$\text{or } PV = RT\left(1+\frac{b}{V}+\frac{b^2}{V^2}+\frac{b^3}{V^3}+\cdots\right) - \frac{a}{V}$$

$$\simeq RT\left(1+\frac{b}{V}\right) - \frac{a}{V} \simeq RT + RT\left(b-\frac{a}{RT}\right)\frac{1}{V}$$

So, the virial form of VDWE as power series of V is

$$Z \simeq 1 + \left(b - \frac{a}{RT}\right)\frac{1}{V}$$

The 2nd virial coefficient, $B = b - \frac{a}{RT}$

Thus, evaluation of B is possible from a and b.

- ✓ $T_B \rightarrow$ **The temperature at which real gases obey ideal gas equation over an appreciable Pr. range**

At the Boyle temp, T_B, $B = 0$, Thus, $b - \frac{a}{RT_B} = 0$ or $T_B = \frac{a}{Rb}$

Therefore, the virial form of the VDWE can produce the expression of T_B.

Real Gases: Explanation of Amagat's Curves in the Light of VDW Eq.

The virial form can easily be expressed as power series of P by replacing $V = \frac{RT}{P}$ (taking approximation for small term).

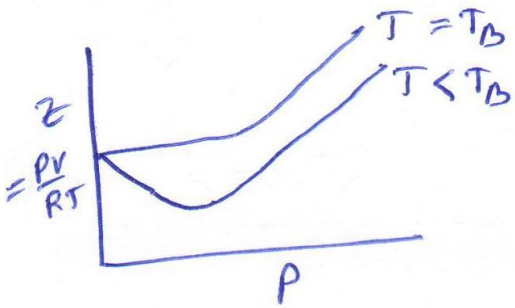
So the form is, $Z = 1 + \frac{1}{RT}\left(b - \frac{a}{RT}\right)P$

A more rigorous mathematical treatment can be exercised to get a form consisting of 3rd virial coefficient. This form is

$$Z = 1 + \frac{1}{RT}\left(b - \frac{a}{RT}\right)P + \frac{a}{(RT)^3}\left(2b - \frac{a}{RT}\right)P^2.$$

It can be shown that the 2nd virial coefficient is $-ve$, while 3rd virial coefficient is $+ve$.

Thus, when P is low, 2nd term dominate and Z decreases with increase of P but when P is high, the 3rd term dominates and Z increases with the increase of P in the Z vs. P diagram.

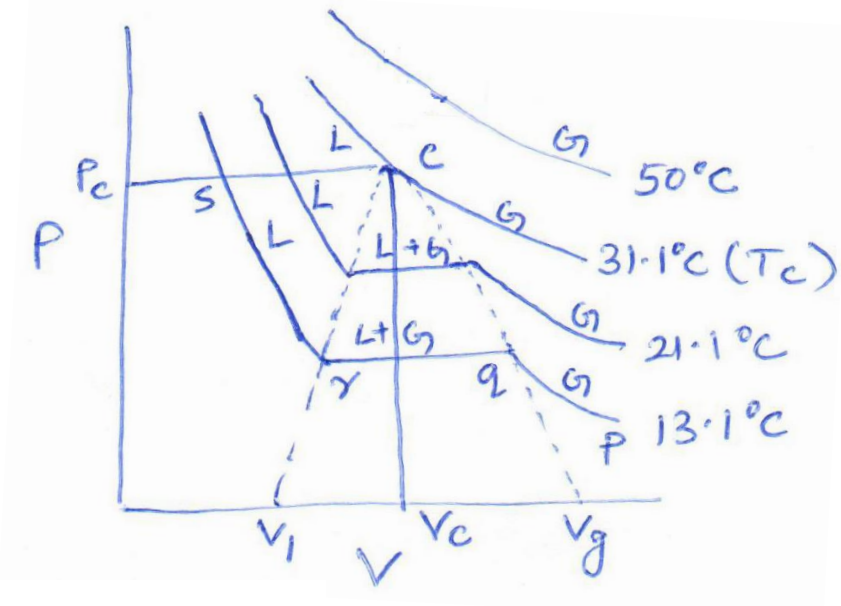


A complete description of Z vs. P curve is obtained by this virial form of VDWE. More accurate form of VDWE is

$$Z = 1 + \frac{1}{RT}\left(b - \frac{a}{RT}\right)P + \frac{a}{(RT)^3}\left(2b - \frac{a}{RT}\right)P^2 + \frac{aP^2}{(RT)^4}\left[3b^2 - \frac{2a^2}{(RT)^2}\right]$$

Questions

Along the critical temp. isotherm, the slope is zero at the critical point (C) and is –ve on either side of the point. Thus, the slope is maximum (zero value is greater than –ve values) at the critical point. This slope is function of V and its derivative w.r.t. V is again zero at the point.



At critical point, the curve has a flat inflexion.

From the properties of curves, at an inflexion, both first and second derivatives are zero.

Thus, the condition of the critical point is given by

$$\left(\frac{\partial P}{\partial V}\right)_T = 0 \quad \text{and} \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0$$